

Factory-to-Customer Shipping Route Efficiency Analysis

A Case Study of Nassau Candy Distributor

Abstract

This project analyzes factory-to-customer shipping route efficiency for Nassau Candy Distributor. The analysis focuses on shipment lead time, route volume, lead-time variability, delay frequency, geographic bottlenecks, and shipping-mode performance. The workflow includes data validation and cleaning, factory assignment from product information, route-level KPI calculation, comparative analysis, and an interactive Streamlit dashboard.

1. Introduction

Efficient shipment routing is important for understanding delivery performance and identifying operational bottlenecks. This study examines historical order and shipment records from Nassau Candy Distributor and converts the available order, shipping, geographic, product, and financial fields into route-level performance indicators.

2. Problem Statement

The objective is to identify shipping routes and geographic areas that exhibit relatively high lead times, high variability, frequent above-average lead times, or high shipment volumes combined with comparatively poor performance. The study also examines whether observed performance differs across shipping modes.

3. Objectives

1. Validate order and ship dates and calculate Shipping Lead Time.
2. Remove invalid or negative lead-time records and handle missing shipment information.
3. Standardize geographic fields.
4. Map products to their corresponding factories.
5. Create factory-to-customer region and state routes.
6. Calculate shipment volume, average lead time, variability, delay frequency, and Route Efficiency Score.
7. Rank routes from faster to slower based on average lead time.
8. Identify geographic bottlenecks and high-volume routes with above-average lead time.
9. Compare shipping modes using lead-time and available financial indicators.
10. Develop an interactive Streamlit dashboard.

4. Dataset and Data Preparation

The source dataset, Nassau Candy Distributor.csv, contains 10,194 records and 18 original columns, including order, shipping, geographic, product, sales, units, gross profit, and cost fields. Dates are stored in DD-MM-YYYY format. The cleaned dataset adds Shipping Lead Time. Product names are mapped to five factory groups: Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory.

5. Methodology

Shipping Lead Time is calculated from Ship Date minus Order Date. Route-level analysis is performed by grouping records by Factory and customer geography. Average Lead Time is the mean shipping lead time and Lead-Time Variability is represented by standard deviation. Delay Frequency is defined in this project as the percentage of shipments whose Shipping Lead Time is above the overall average lead time. Route Efficiency Score = $(100 / \text{Average Lead Time}) \times 1000$. Higher values indicate shorter average lead times. This is a project-specific index, not an industry standard.

6. Route Efficiency Results

The overall average Shipping Lead Time is approximately 1,320.84 days. Observed values range from approximately 904 to 1,642 days and are retained as supplied by the dataset. At the factory-to-region level, Sugar Shack to Gulf has the lowest observed average lead time at 1,091.25 days across 4 shipments, while Sugar Shack to Pacific has the highest at 1,516.67 days across 3 shipments. These low-volume routes should be interpreted cautiously. High-volume above-average routes include Lot's O' Nuts to New York (645 shipments; approximately 1,330.52 days), Lot's O' Nuts to Washington (269; approximately 1,343.96 days), Wicked Choccy's to Washington (221; approximately 1,375.72 days), and Lot's O' Nuts to Tennessee (109; approximately 1,390.96 days).

7. Geographic Bottlenecks

State-level analysis considered states with at least 50 shipments. Tennessee has approximately 1,391.49 days average lead time across 183 shipments, Indiana approximately 1,381.50 days across 149 shipments, and Washington approximately 1,360.66 days across 506 shipments. These are descriptive observations and do not establish the causes of the observed performance.

8. Ship Mode Comparison

The four shipping modes are First Class, Same Day, Second Class, and Standard Class. Standard Class has 6,120 shipments and an average lead time of approximately 1,314.33 days. First Class has 1,548 shipments at approximately 1,338.28 days; Same Day has 547 at approximately 1,333.44 days; and Second Class has 1,979 at approximately 1,323.85 days. Grouped expedited modes (First Class, Same Day, Second Class) average approximately 1,330.62 days versus 1,314.33 days for Standard Class. This is an observed dataset result,

not an industry-wide conclusion. The dataset has no explicit shipping-cost field, so Sales and Gross Profit are only available financial indicators.

9. Key Performance Indicators

The dashboard uses Total Shipments, Average Lead Time, Route Volume, Delay Frequency (above overall average), Route Efficiency Score, and Lead-Time Variability. Filters include factory, state/province, ship mode, order date range, and minimum shipment volume.

10. Streamlit Dashboard

An interactive Streamlit dashboard was developed with route efficiency analysis, route-level KPI data, geographic shipping analysis, ship-mode comparison, and route drill-down capabilities. The application is publicly deployed for interactive exploration.

11. Limitations

The dataset contains unusually large shipping lead-time values, so the source observations are retained and reported transparently. There is no explicit shipping-cost field. The Route Efficiency Score is a custom project index. Low-volume routes can produce unstable averages. The analysis is descriptive and does not establish causal reasons for delays.

12. Conclusion

This project provides a structured framework for evaluating factory-to-customer shipping route performance using historical shipment data. Combining data cleaning, factory mapping, route KPIs, geographic analysis, ship-mode comparison, and an interactive dashboard highlights areas for further operational investigation. Future work could incorporate transportation cost, carrier, SLA, distance, warehouse processing time, and delivery exception data.

References

1. Nassau Candy Distributor dataset supplied for the project.
2. Streamlit documentation: <https://docs.streamlit.io/>
3. Pandas documentation: <https://pandas.pydata.org/docs/>
4. Plotly Python documentation: <https://plotly.com/python/>